KLE Technological University | Creating Value, Leveraging Knowledge

KLE TECH HACKFEST 2026

MENTORED BY ALUMNI | POWERED BY STUDENTS

DOMAINS

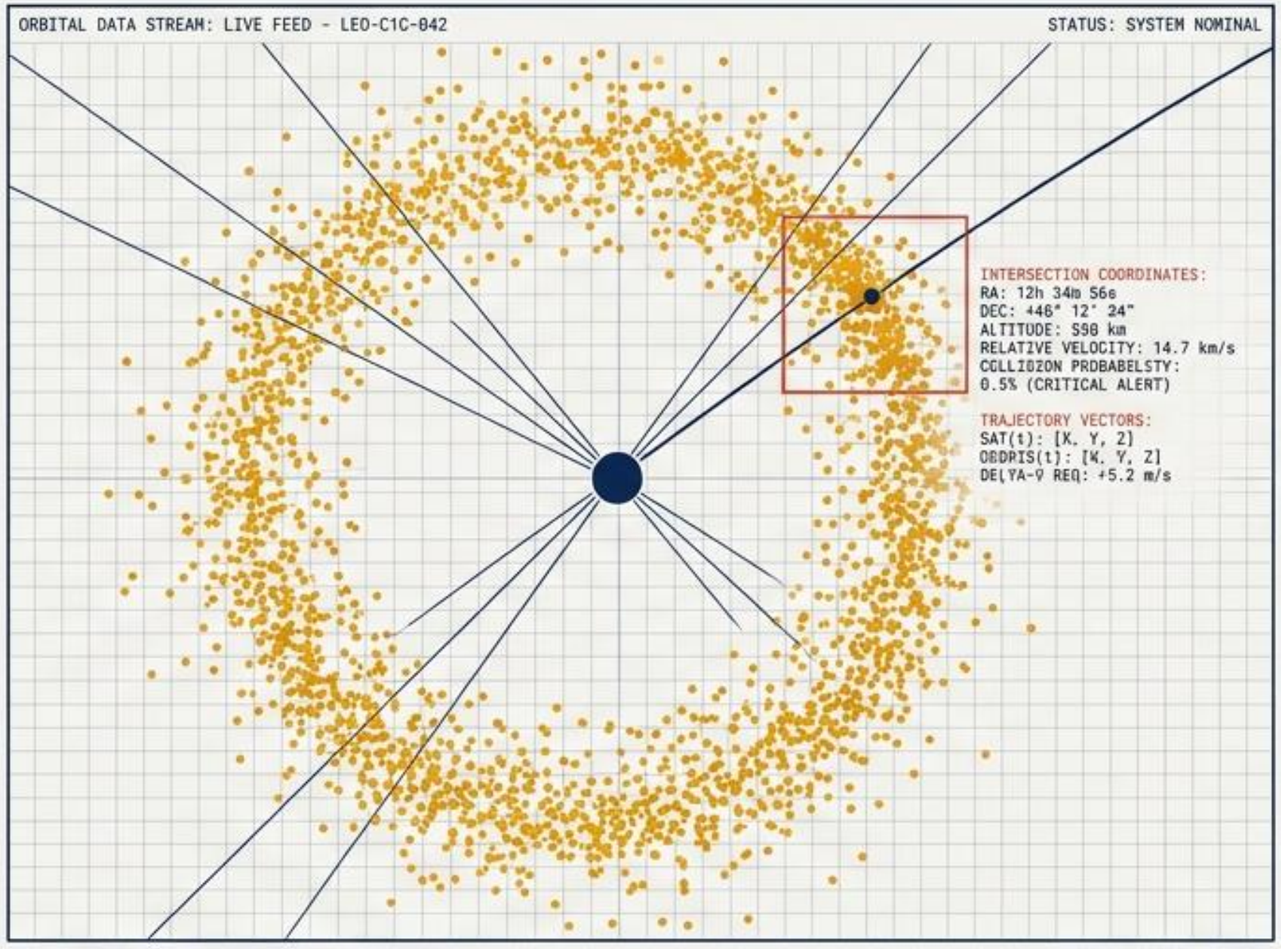
1. DEFENSE, AEROSPACE, NAVY, GOVERNMENT & SOVEREIGN AI
2. AI DATA, AI-WORKLOADS STORAGE, DATA SECURITY, THREAT INTELLIGENCE
3. AI-FACTORY, MINING AI, HI-TECH, DIGITAL TWINS, EDGE AI & ROBOTS
4. PRODUCT MANAGEMENT
5. CONTEXT-AWARE APPLICATION AGENT & DYNAMIC PRICING ENGINE
6. AI IN HEALTHCARE

TOTAL PRIZE MONEY ₹1,75,000

9th | 10th | 11th OCTOBER 2026

INTERNSHIP OPPORTUNITY

HACKATHON CHALLENGE 1C:
SPACE DEBRIS RISK PREDICTION



MISSION:

Automating the Go / No-Go Collision Decision in Low Earth Orbit.

TARGET:

Build an AI that parses noisy orbital physics, delivers calibrated confidence, and recommends fuel-efficient avoidance maneuvers.

TIME-TO-COLLISION: T-MINUS 00:04:38

KLE TECH HACKFEST 2026
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SPACE DEBRIS CRISIS: UNDERSTANDING THE ORBITAL HAZARD & HOW AI CAN SAVE OUR SATELLITES

THE PROBLEM — A CONGESTED & DANGEROUS HIGHWAY



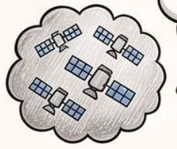


LOW EARTH ORBIT (LEO) TRAFFIC JAM

ACTIVE SATELLITES IN LEO



800 (2019)



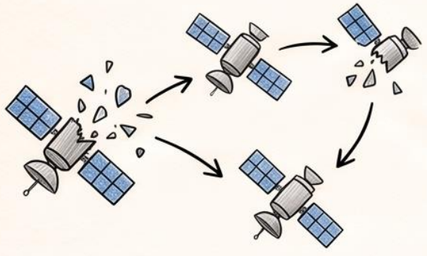
OVER 5,000 (2022)

58,000 PROJECTED (2030)



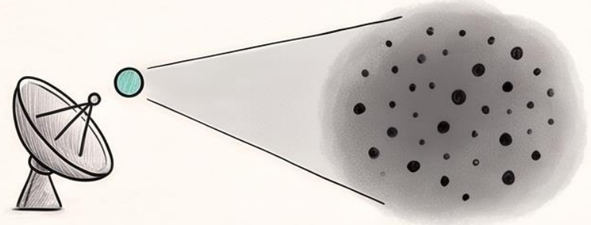
TINY FRAGMENTS CARRY EXPLOSIVE FORCE

1 cm wide hits with hand grenade energy @ 7.5 km/s



THE KESSLER SYNDROME THREAT

Runaway collision chain reaction locks humans out of key orbits



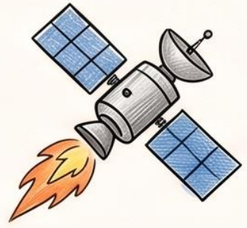
TRACKED: >29,000 (>10 cm)

UNTRACKED: >500,000 (>1 cm INVISIBLE HAZARDS)

THE BOTTLENECK — ALERT FATIGUE & UNCERTAINTY

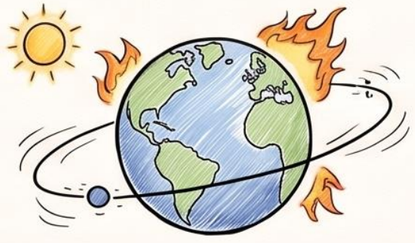


CONJUNCTION DATA MESSAGES (CDMs)



MOVING A SATELLITE HAS REAL COSTS

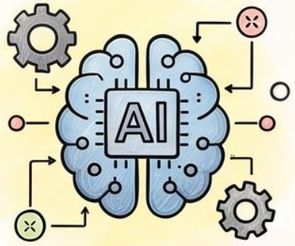
Burns fuel, reduces lifetime, causes downtime

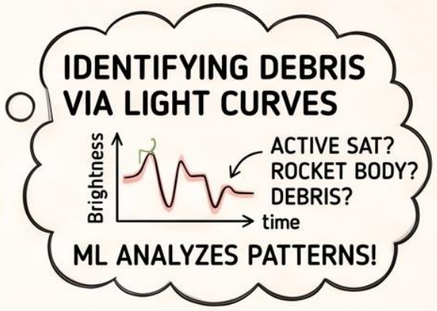


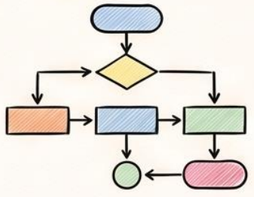
UNPREDICTABLE ATMOSPHERIC DRAG

Solar storms distort position, growing errors into kilometers in days

THE AI SOLUTION — SMART DECISION-MAKING & TRACKING







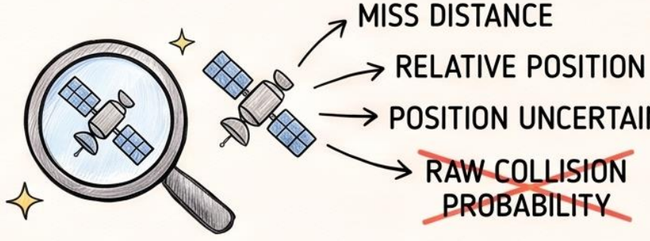
AUTOMATING THE "GO / NO-GO" DECISION

Deep Learning (LSTM) models evaluate close approaches near critical collision thresholds



72% ACCURACY

AI models match expert decisions in high-risk tests (GSOC)



WHAT THE AI PRIORITIZES MOST

Drastically lowers false alarms by focusing on key features

From European Space Agency:

https://www.esa.int/ESA_Multimedia/Videos/2025/04/Space_Debris_Is_it_a_Crisis

https://dlmultimedia.esa.int/download/public/videos/2026/02/027/2602_027_AR_EN.mp4

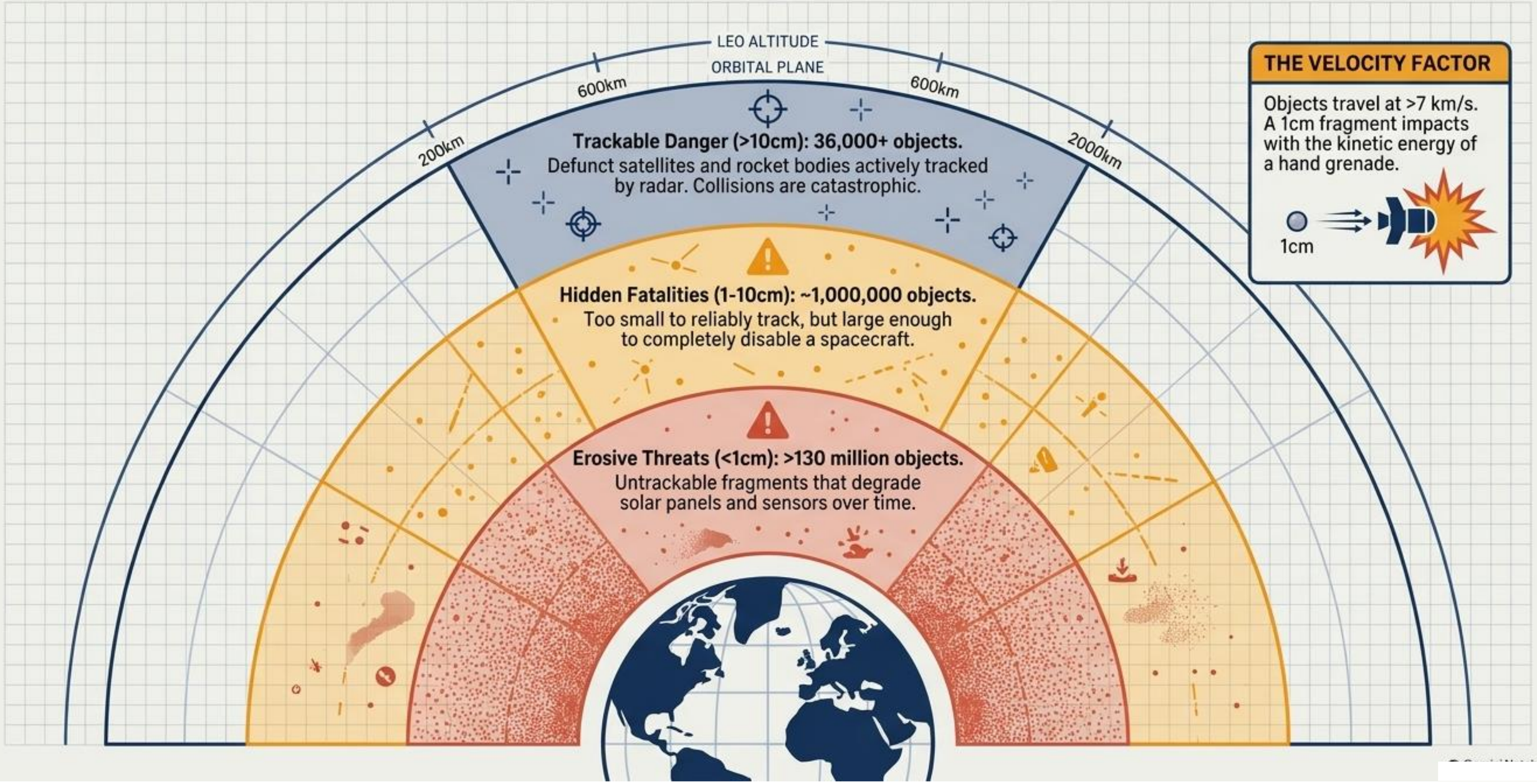
USER PERSONAS FOR SPACE DEBRIS MANAGEMENT

Who faces the risk, and what they need from the solution

PERSONA	MAIN CONCERN	SUPPORT NEEDED
Satellite operations engineer	Collision alerts compete with limited fuel and mission uptime.	Timely go / no-go guidance and fuel-efficient maneuver options.
Orbital risk analyst	Uncertain tracking data can change collision-risk estimates.	Traceable risk estimates, clear confidence levels and alert priority.
Spacecraft mission designer	Design choices shape debris exposure and end-of-life risk.	Orbit-risk comparisons and disposal planning from the start.
Space safety regulator	Debris growth threatens safe access to shared orbital space.	Evidence of mitigation compliance and transparent risk reporting.

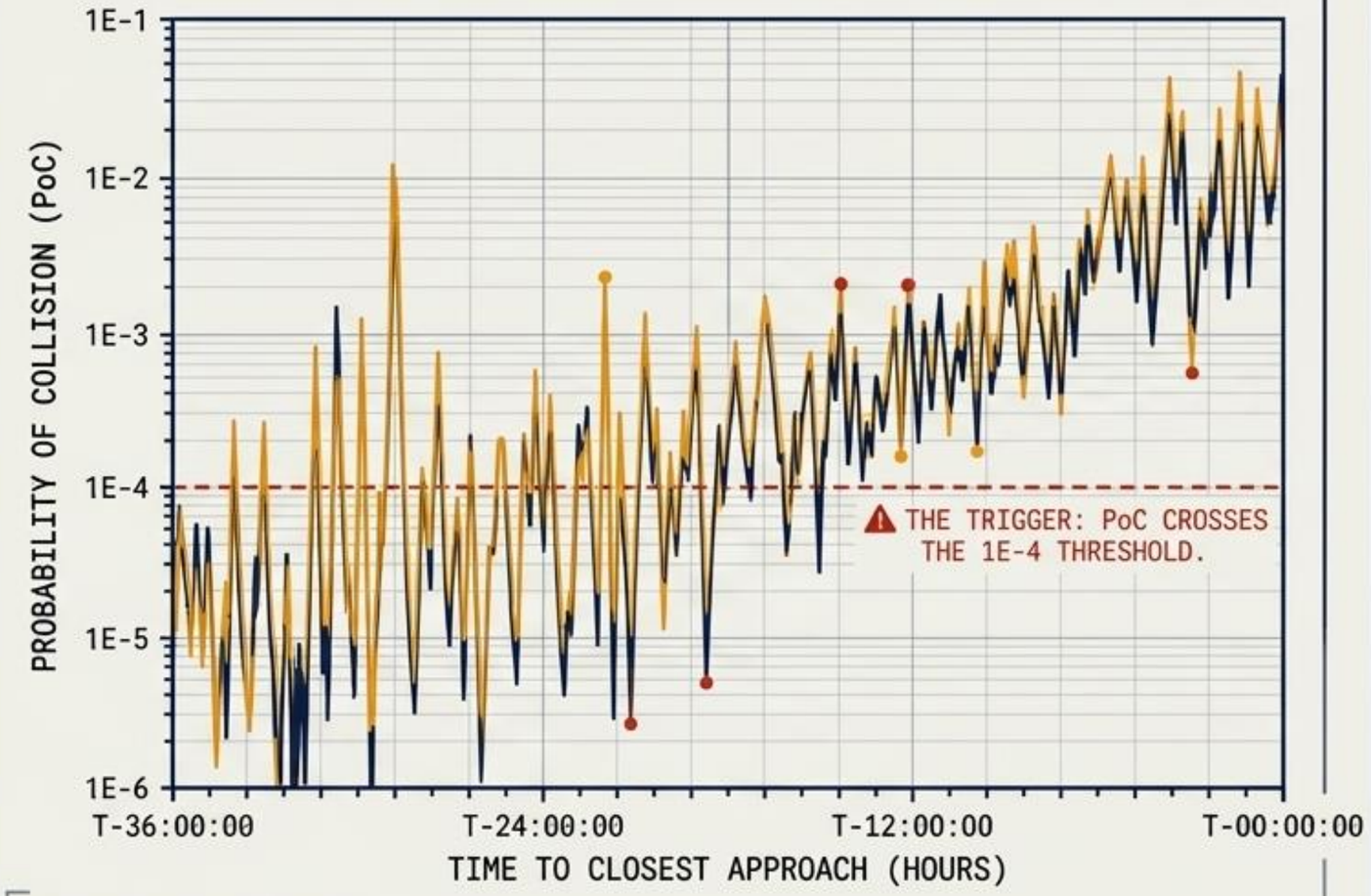
Illustrative role-based personas informed by NASA and ESA guidance

THE PHYSICAL REALITY OF LOW EARTH ORBIT



THE 02:00 AM OPERATOR'S DILEMMA

CONJUNCTION DATA MESSAGE (CDM) TELEMETRY



CDM ID: 2024-09-15-012
TCA: 2024-09-17T02:00:00Z
MISS DISTANCE: 1.2 km
DELTA-V: 0.5 m/s
CONFIDENCE LEVEL: LOW -> HIGH

OPERATOR DECISION MATRIX

	ACTION TAKEN (MANEUVER)	NO ACTION TAKEN (HOLD)
	FALSE POSITIVE ⚠	FALSE NEGATIVE ⚠
	• BURNED DELTA-V (FUEL) • SHORTENED MISSION LIFE • INTERRUPTED SCIENCE/COMMUNICATIONS • (OPERATORS RECEIVE HUNDREDS OF ALERTS WEEKLY; 99% ARE NOISE).	• $500M ASSET DESTROYED • SECONDARY DEBRIS CLOUD GENERATED (KESSLER CASCADE)

THE PHYSICS BOTTLENECKS HINDERING PREDICTION

ATMOSPHERIC DRAG

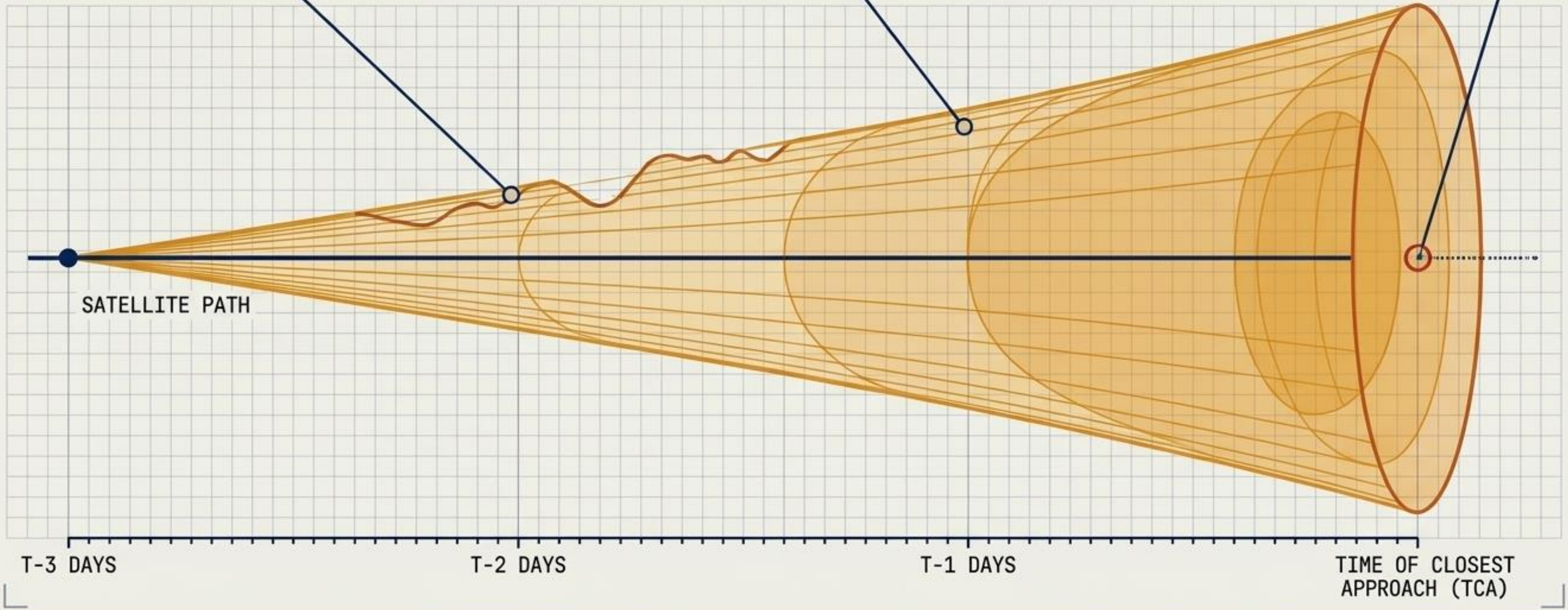
The largest error source in LEO. Solar storms (F10.7, Kp/Ap indices) rapidly alter atmospheric density, decaying orbits unpredictably.

UNREALISTIC COVARIANCES

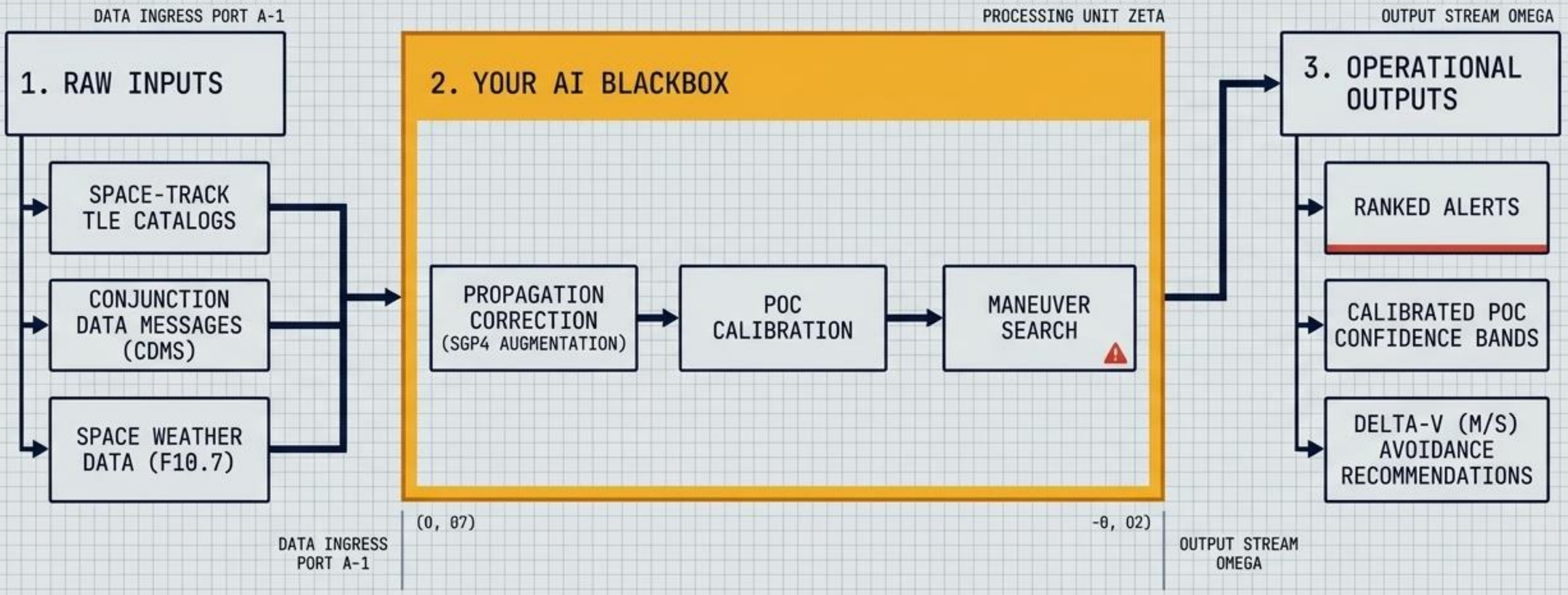
CDM uncertainty bubbles are often miscalibrated. PoC math breaks when covariance is artificially too small or too large.

THE CLASS IMBALANCE TRAP

Real collisions are incredibly rare. Naive machine learning models achieve 99.9% accuracy by simply predicting nothing will happen.



CHALLENGE 1C FUNCTIONAL ARCHITECTURE



MISSION CONTROL CENTER ⚠ OPERATIONAL STATUS: CRITICAL 2024-09-15 02:00:00 UTC

STATE-OF-THE-ART AI ARCHETYPES FOR ORBIT PREDICTION

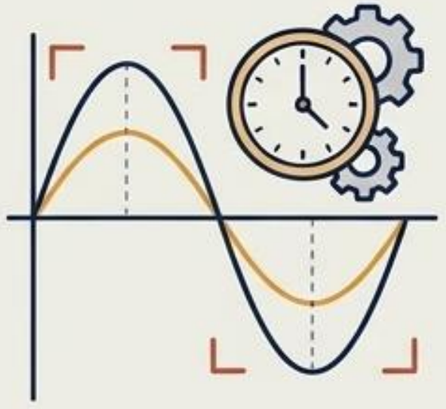
STACKED ENSEMBLES (XGBOOST + DNN)



Best for calibrating probabilties.

Overcomes extreme class imbalance by mapping Monte Carlo simulated outcomes onto gradient boosted trees to refine PoC.

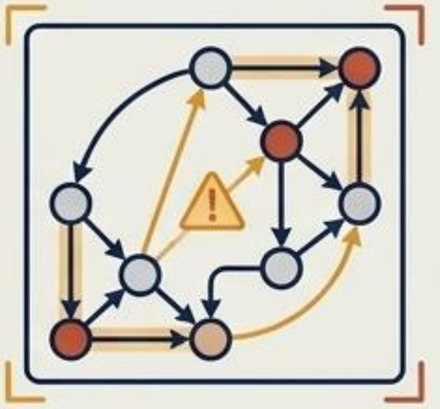
TIME SERIES DEEP LEARNING (LSTM)



Best for sequential CDMs.

Learns the temporal pattern of an approaching threat over several days rather than relying on a single snapshot.

REINFORCEMENT LEARNING (DRQN/POMDP)

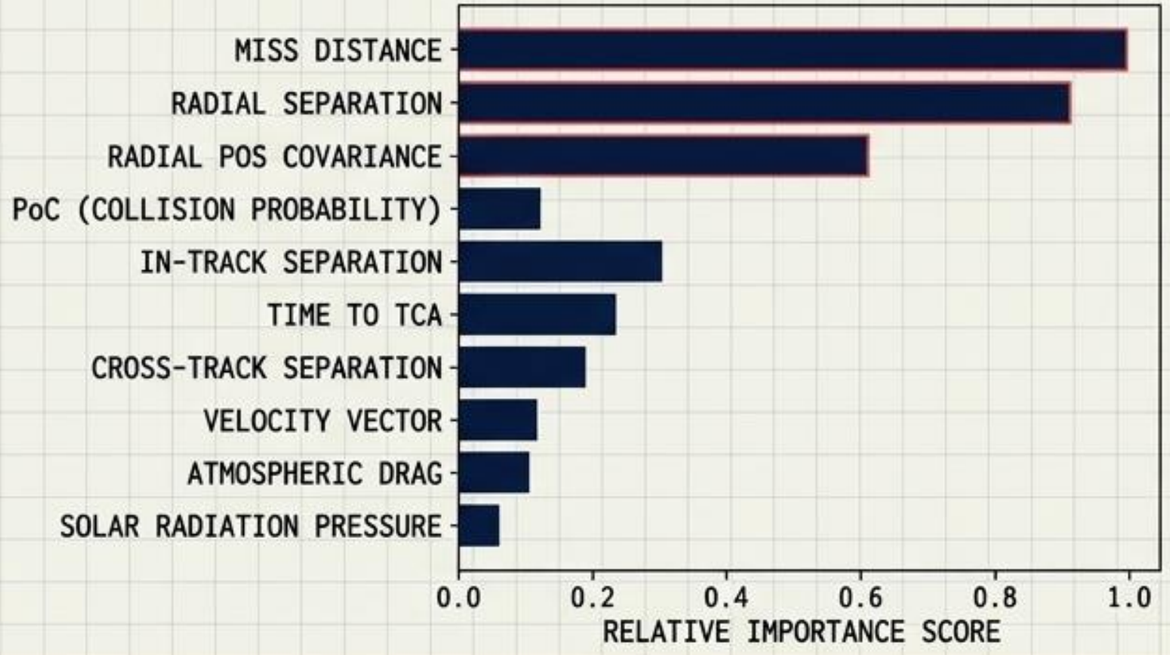


Best for maneuver optimization.

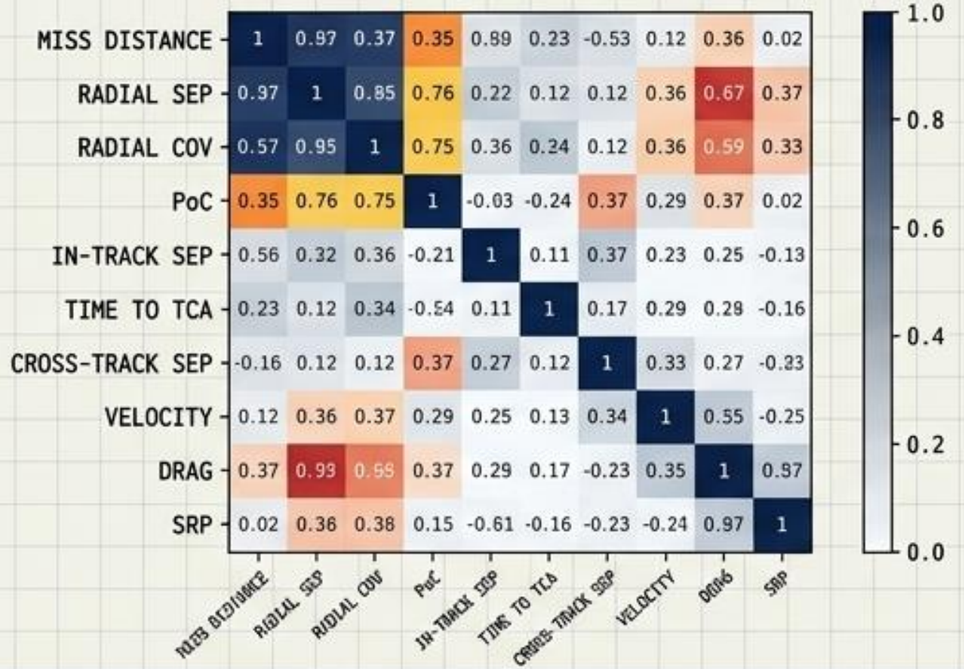
Treats the noisy, unpredictable CDM data as a Partially Observable environment to find the cheapest fuel burn.

DECODING THE CONJUNCTION DATA FEATURES

LSTM FEATURE IMPORTANCE



FEATURE AUTOCORRELATION MATRIX



⚠ SUMMARY ANALYSIS

THE COUNTERINTUITIVE TRUTH:

AI models weight specific geometric features higher than the raw Probability of Collision (PoC) when replicating operator decisions.

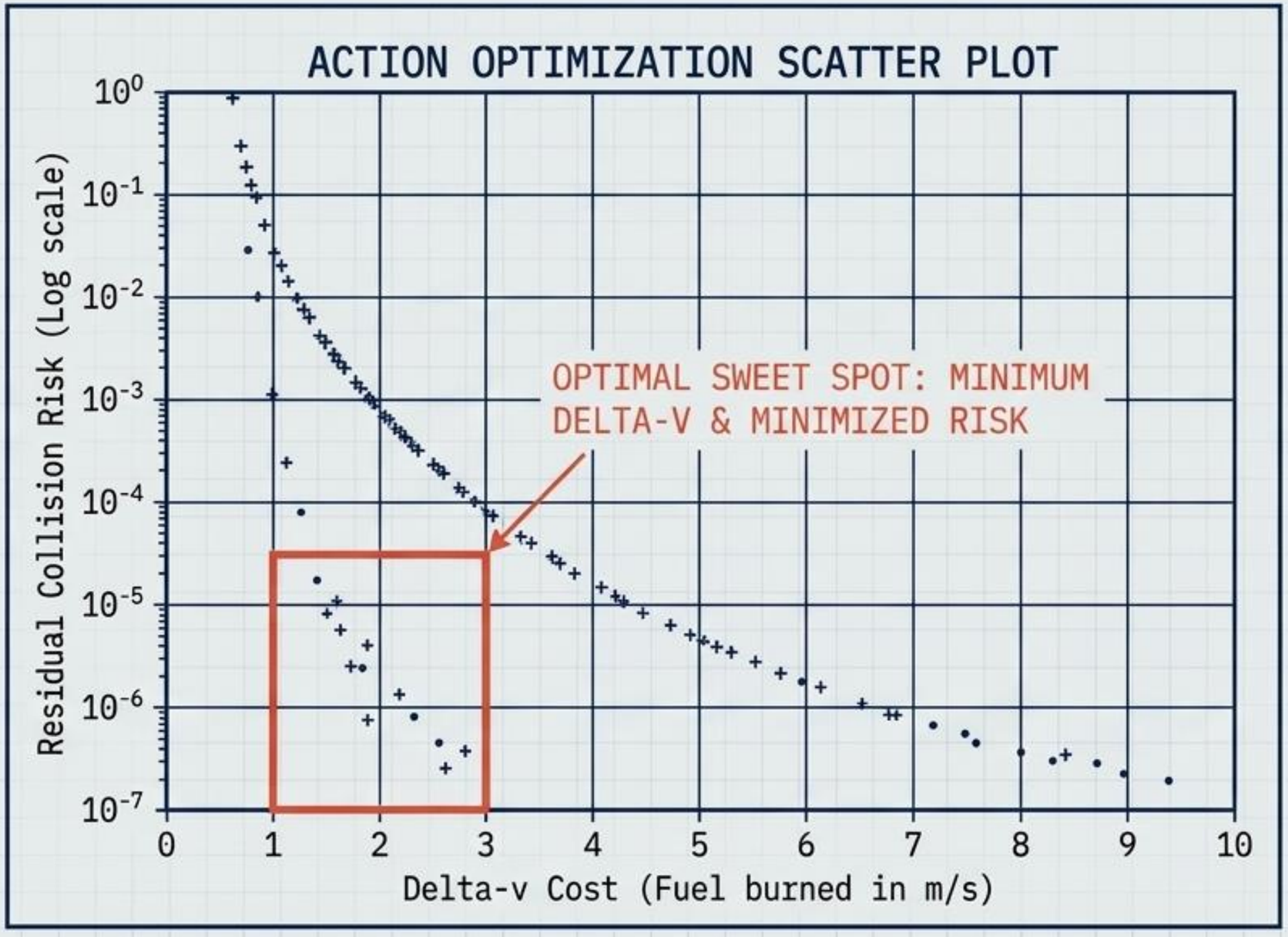
TOP DRIVERS:

Miss Distance, Radial Separation, and Radial Position Covariance of the secondary object.

TAKEAWAY:

Your model must understand the geometry of the miss, not just the headline probability number.

AUTOMATING THE COLLISION AVOIDANCE MANEUVER (CAM)



THE OPTIMIZATION GOAL:

Move beyond predicting collisions to recommending the exact physical maneuver required to survive.

THE TRADE-OFF:

Balance risk reduction against the absolute minimum delta-v penalty.

CONSTRAINT:

The avoidance burn must not place the satellite into the path of a different debris object.

EXPLAINABILITY (XAI) AND SPACE TRAFFIC MANAGEMENT

THE TRUST DEFICIT:

Operators will not blindly execute a maneuver on a $500M asset based on a black-box output.

[ASSET VALUATION: $500M+]
[RISK ASSESSMENT: UNVERIFIED]

--

[LAST ACTION: 11.30%]
[RISK ASSESSMENT: UNVERIFIED]

XAI INTEGRATION (SHAP/LIME):

Your output must explain WHY it triggered an alert, detailing the driving parameters behind the decision.

SHAP EXPLANATION MODEL: DATA TREAK STREAMING: ... RADIAL COV: +30% MISS DISTANCE: -20%

LIME INTERPRETER: DRIVE STREAMING FEATURES: ... DRAG MODEL: +15% DRAG MODEL: +15%

⚠ EXAMPLE TELEMETRY READOUT:

[ALERT]: Radial covariance decreased 40% in last pass. ⚠
[DIAGNOSTIC]: Drag model indicates secondary orbit decay intersecting in 31hrs.
[ACTION]: Recommend 0.5 m/s along-track burn. 🔥

MISSION CONTROL CENTER OPERATIONAL STATUS: CRITICAL ⚠ 2024-09-15 02:00:

TARGET SPECS & JUDGING KPIS



Brier Score (Calibration)

Does a 1-in-10,000 predicted risk actually happen 1-in-10,000 times?



Hit/Miss Rate

>95% recall on high-risk events (never miss a true threat) while dropping false alarm rates.



Lead Time

Flagging genuine risks 3 to 7 days out from Time of Closest Approach.



Throughput

Ability to screen the full object catalog in minutes, not hours.

> SYSTEM READY. AWAITING INPUT.